Jiahang Bai

AI Undergraduate at Jilin University · Embodied Foundation Models

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Education

Jilin University

Undergraduate in Artificial Intelligence · School of Artificial Intelligence *2024–Present*

GPA 3.63 / 4.00

Selected coursework Probability Theory and Mathematical Statistics A (94/100), Deep Learning (91.1/100), Machine Learning and Pattern Recognition II (92/100), Frontiers of Artificial Intelligence I (96/100), Data Structures and Algorithm Analysis II (91/100), and Evolutionary Computation (90.6/100)

Research Interests

How can embodied foundation models learn to use longer interaction histories? I am interested in pretraining for embodied reasoning and sequence modeling, and in action token representations that translate multimodal reasoning from models such as GPT Astra into robot control

Research and Industry Experience

Dexmal

Embodied AI Algorithm Intern | *Mentor: Ze Chen* *Jan 2026–Present*

Embodied Reasoning Pretraining | **100-GPU-Scale Training** | **3 Papers Under Review (1 as First Author)** | **Real-Robot Tests**

- **Jan–Feb · π_0 Sampling Distillation** Explored denoising-step distillation to accelerate action generation
- **Mar–May · DM0.5 Embodied Reasoning Pretraining** Designed embodied chain-of-thought (ECoT) tasks and specified candidate data and supervision for spatial grounding, task progress, and hierarchical planning
- **Jun–Jul · VLA Context Modeling** Explored history selection and readout for DM05 Memory
  Team result · DM05-MEM later ranked first on RoboDojo-Sim (Aug 2026)
- **Aug–Present · Algorithm Research** Contributed to three papers under review on memory, tactile VLA, and online RL. Achieved SOTA among compared methods on RoboTwin-MeM (+20.1 pp vs EventVLA), RMBench (+12.9 pp vs EventVLA), UniVTAC (+16.84 pp vs FTP-1), and ManiSkill-Fine (+11.27 pp on seen and +8.54 pp on unseen initializations vs RLT). Ran real-robot evaluations on DOS-W1, Mobile ALOHA, RealMan, and ARX and supported training on 100-GPU-scale clusters

HKUST (Guangzhou)

Research Intern (Remote) | *Mentors: Qiwei Liang and Renjing Xu* *Jul–Aug 2026*

- **Multi-Task RL for World Action Models** Built a training prototype and explored using forward and inverse dynamics supervision from rollouts to refine visual–action representations for policy updates

Jilin University

Undergraduate Researcher | *Mentor: Yixing Gao* *Dec 2025–Present*

- Reproduced OpenVLA and $\pi_{0.5}$, fine-tuned shirt-folding policies and explored retries after failure, with experiments on Baxter, RealMan, and LimX TRON 2
  Built AdaChunkWAM to adapt FastWAM's action chunks and replanning to predicted scene changes

CAS Undergraduate Innovation and Research Training Program

Research Intern (Remote) | *Mentor: Jianghao Wang* *Dec 2024–Jun 2025*

- **LLM Weak Supervision** Designed and refined LLM prompts to label social media text for sentiment and depression-related signals, and explored fine-tuning BERT on generated labels

Selected Papers

Under review

S²Mem: From Sparse Cues to Decision-Relevant Memory in Vision-Language-Action Models

Sole first author · **Jiahang Bai**[1], Leyang Wang[2], Zixuan Huang[2], Wenxuan Guo[3], Xutao Sun[1], Kaixi Ying[1], Xiaowen Qin[1], Ze Chen[2], Erjin Zhou[2], Yixing Gao[1]

[1] Jilin University · [2] Dexmal · [3] Tsinghua University

Key cues in long-horizon manipulation can disappear quickly, and the current view suffices for most demonstrated actions, giving policies little supervision for using history

As frames leave the short-term window, S²Mem uses subsequent robot motion to decide which to save. An event-aligned prediction objective trains memory queries to recover frozen DINOv2 features of masked events. LoRA adapts the language model while the pretrained VLM backbone stays frozen

Success rates reached 95.3% on RoboTwin-MeM and 80.7% on RMBench (+20.1 and +12.9 pp over EventVLA), and 83.33% on Mobile ALOHA (+35.0 pp over the best baseline)

FutureTacVLA: Unifying Interaction Foresight and Contact Dynamics Awareness for Visuotactile Manipulation

Co-author · Leyang Wang[1], Yi Wang[2], **Jiahang Bai**[3], Zixuan Huang[3], Peili Ma[4], Yongda Chen[3], HuaZhihua[3], Ruitao Zhang[3], Ze Chen[3], Erjin Zhou[3], Jikai Wang[1]

[1] University of Science and Technology of China · [2] Beijing Institute of Technology · [3] Dexmal · [4] Shantou University

Visuotactile forecasting and contact modeling · vs FTP-1, UniVTAC 83.50% (+16.84 pp), real robots 77.3% (+25.3 pp)

Sight-RLT: Control-Oriented RL Tokens from Vision-Language-Action Models

Co-author · Zixuan Huang[1], Yanxiang Ma[2], Jianyuan Guo[1], Jiaqi Liu[2], Leyang Wang[3], **Jiahang Bai**[3], Ze Chen[3], Kaixin Liu[3], Erjin Zhou[3], Chang Xu[2]

[1] City University of Hong Kong · [2] University of Sydney · [3] Dexmal

Frozen VLA features for online RL · ManiSkill-Fine vs RLT (+11.27 pp seen / +8.54 pp unseen)

Awards and Honors

RoboCup China Open · Humanoid Soccer (Large)

Team Captain | *National Second Prize and Top 8 Nationwide* *2026*

- Designed and developed core algorithms for robot positioning and return-to-field behavior, then used feedback from real-robot trials to tune movement speed and improve shooting performance
- Coordinated the team, organized practice and task assignments, and served as the point of contact with competition organizers

Mathematical Contest in Modeling

Team Captain | *Jilin University's sole Problem C Finalist* *2025*

- Led problem decomposition and model formulation for Problem C and designed and implemented core algorithms for Olympic gold-medal forecasting and estimation of the great-coach effect, using LSTM-based models for medal forecasting

Additional Awards Second-Class Scholarship, National College Student Mathematics Competition (Third Prize), Jilin Provincial Virtual Reality Competition (Second Prize), and SuperMap Cup University GIS Competition (Merit Award)

University Leadership

Class Monitor and Student Counselor Assistant, Jilin University